

Aliah University

End Term Examination Spring Semester 2024

(For 2nd Year 3rd Semester of BCA)

Subject Name: Microprocessor

Subject Code: ECEUGOE02

Full Marks: 80

Time: 3 hours

Group-A

Answer Any Five Questions (5x2=10)

1. a) In which mode all ports of 8255 functions as simple I/O? what is opcode?
- b) Explain the function of pin ALE and WR' in 8085 microprocessors.
- c) List the special registers available in 8085. Mention size of each of them.
- d) What is the addressing mode of LDA 2234H? Justify your answer.
- e) Explain the operation of STAX D instruction of 8085.
- f) What is the function of RESET pin of 8086 PPI?
- g) Why the size of SP is 16 bits? Justify your answer.

Group-B

Answer Any Four Questions (4x5=20)

2. Explain with example, the difference between maskable and non-maskable interrupt.
3. With suitable example explain any two addressing modes of 8085.
4. Explain PUSH and POP instruction with example.
5. Define instruction cycle and Machine Cycle? Draw the timing diagram of opcode fetch. 2 + 3
6. Write a program to subtract two numbers located at memory locations 2000 and 2001 respectively and stored the result in memory location 2002. Show the contents of registers and memory locations used in your program.
7. Explain arithmetic and logical instruction with at least one example from each category. Why transfer instructions are also known as copy instruction? 3 + 2

Group-C

Answer Any Five Questions (5x10=50)

8. Draw the block diagram of programmable Interrupt controller (8259) and explain its operations. 10
9. a) With neat diagram explain detailed architecture of 8086. 8 + 2
- b) Show how pipeline is available in 8086.
10. a) What is flag? What is usage of flags? What are the different flags available in 8085? What is the size of flag registers? 1 + 2 + 1 + 1
- b) Specify the content of registers A and B and the status of flags S, Z, Cy when the following instructions are executed. (i) ORI A, E8H (ii) ADI B, 4AH 5

11. Calculate the physical address, when segment address is 1085H and effective address 4537. Explain the register organization of 8085. What are the uses of AD0 to AD15? What are the functions of status pins in 8086? $2+3+1+4$

12. What are the different operation modes available in 8255? Explain BSR mode of 8255 with example. Write control word to set PC3 and reset PC6 in 8255. What is the maximum size of memory that can be interfaced with 8085? $2+4+2+2$

13. List the advantages and disadvantages of partial decoding. Can we use Isolated I/O for partial decoding? Justify your answer. Interface an 2KB EPROM chip with an 8085 microprocessor with starting address E000H? Draw the interfacing diagram. $2+2+6$

~~Screen~~, CRM.

i) create identity - i)

~~billing~~

ii) hotel voucher

iii) ~~voucher~~ billing.

Aj kamaenga kal baith
kar khayenga apne kismat
par bharosa rakn. --- Pyar
to dubara bhi ho jayenga ---
Haa --- meny --- Jaan ---